

Power Optimization: Idle Draw Baseline

Documented October 2026

Overview

This entry records the first power-focused pass on the Proxmox host (HP EliteDesk, i5-6600), measured with turbostat and PowerTOP. It covers three states: GTX 960 installed, GTX 960 removed, and GTX 960 removed with the pfSense VM shut down. This isn't a full project. It's a landing pad for small efficiency changes to the homelab regarding power, and it'll get updated as new tweaks are tested.

Goal

The goal is for the host to run 24/7. To plan this in best practice, I wanted to record a baseline for some of the actual idle power usage so I could get a solid understanding of the amount of electricity I am using. Some other goals are listed below:

- Decide whether keeping the GTX 960 in the box for the planned Jellyfin passthrough is worth its idle cost
- Find out what keeps the CPU package out of deep sleep states
- Identify wakeup sources worth fixing

The USB RTL8153 stays plugged in permanently, since it is pfSense's LAN interface. Anything that depends on removing it is out of scope.

Method

Every run used the same command, sampling every 5 seconds for 20 samples:

```
turbostat --Summary --quiet --show  
Pkg%pc2,Pkg%pc3,Pkg%pc6,Pkg%pc7,CPU%c1,CPU%c6,CPU%c7,Bzy_MHz,PkgWatt --interval 5
```

The last row of each run is a partial interval from Ctrl+C and was ignored. PowerTOP was run in interactive mode only. I deliberately did not use --auto-tune, since it can power-manage the USB NIC and the external SSD (which holds the Proxmox root) into dropouts.

Note that PkgWatt is the CPU package power reported by the processor itself. It does not include the GPU's own board power, the SSD, the NICs or PSU losses, so it isn't wall power. An Electric-Measures wall plug is one of the primary components I'd like to add next.

Results

State	PkgWatt	Core C7	Pkg PC2	Pkg PC3/PC6/PC7	
GTX 960 installed, pfSense on	~11.1 W	~65–72%	~53–68%	0.00	
GTX 960 removed, pfSense on	~4.7 W	~72%	~70%	0.00	
GTX 960 removed, pfSense off	~1.1 W	~96–98%	~93–97%	0.00	

Findings

Removing the GTX 960 dropped package power by roughly 6.4 W. Core idle residency barely changed, so the cores themselves were never the issue.

Shutting down the pfSense VM dropped it by a further ~3.5 W. With the VM running, the cores spend about 19% of their time in C6 and 5–6% in C1 instead of sitting almost entirely in C7. That is constant waking from timer ticks, virtual NIC interrupts, and packet handling for the USB LAN NIC.

In all three states the package never reaches PC3, PC6 or PC7. It stays at PC2 even when the cores are 96–98% asleep. That points to something in the platform, not a workload, since a quiet host with only the Pi-hole container still can't get deeper.

PowerTOP with the VM off showed ~457 wakeups per second and 7.6% CPU use. The notable sources:

- tick_nohz_handler (~292/s): the normal kernel timer tick, nothing to fix
- usb_giveback_urb_bh (~32/s): USB transfer completions from the NIC and/or external SSD, which keeps the USB controller from ever being idle
- fail2ban-server (~9/s): busier than expected, likely polling log files
- pve-firewall: spawns ebtables-save and sh processes every cycle, even though filtering is handled by pfSense and Tailscale ACLs

Nuances and caveats

- The pre-pull baseline was captured the day before, on a different boot. Busy MHz was ~900 then versus ~3850 after, and I did not record the CPU governor either time. Some of the 6.4 W difference could come from frequency or boot state rather than the card alone, so treat it as an estimate until it is repeated under identical conditions.
- Because PkgWatt excludes the GPU's own board power, the real saving at the wall from removing the card is likely different from the 6.4 W figure. A wall power meter is needed for the true number, as I mentioned earlier.
- Busy MHz is not an idle clock. At 96–98% C7 it only reflects the rare wakeups that hit turbo.
- Each state was measured once. [Confirm: LXC containers (Pi-hole, Tailscale) were running unchanged in all three runs.]
- The PC2 cap is unresolved. The USB isolation test was skipped because the RTL8153 is staying.

Planned next steps

- Record the CPU governor and rerun the GPU-installed baseline under identical conditions
- Check the BIOS for the package C-state limit (set to C10 or auto) and any ASPM or PCIe power-saving option
- Review ASPM status with `lspci -vv`, plus the PowerTOP Tunables tab, avoiding the USB and SSD items
- Set `kern.hz=100` in the pfSense guest to cut timer wakeups
- Switch `fail2ban` to `backend = systemd`
- Evaluate whether `pve-firewall` can be disabled
- Plug-in wall power meter reading for a full-system idle number
- Decide on GPU placement: leave it out until the Jellyfin passthrough project starts

Screenshots

- turbostat: GTX 960 installed, pfSense on

```
root@proxmox:~# turbostat --Summary --quiet --show Pkg%pc2,Pkg%pc3,Pkg%pc6,Pkg%pc7,CPU%c1,CPU%c6,CPU%c7,Bzy_MHz,PkgWatt --interval 5
Bzy_MHz CPU%c1 CPU%c6 CPU%c7 Pkg%pc2 Pkg%pc3 Pkg%pc6 Pkg%pc7 PkgWatt
900      3.91  20.27  67.46  61.30  0.00  0.00  0.00  11.10
910      3.75  18.33  72.42  68.42  0.00  0.00  0.00  11.03
900      6.74  20.05  59.95  52.70  0.00  0.00  0.00  11.30
900      4.51  19.55  67.48  61.79  0.00  0.00  0.00  11.12
900      4.27  19.69  68.50  62.89  0.00  0.00  0.00  11.11
1017     3.95  19.17  69.81  65.34  0.00  0.00  0.00  11.15
1055     3.86  19.90  65.26  55.96  0.00  0.00  0.00  11.44
900      3.87  18.38  71.98  67.76  0.00  0.00  0.00  11.05
955      3.99  19.31  69.44  63.84  0.00  0.00  0.00  11.19
909      3.86  18.93  70.75  65.62  0.00  0.00  0.00  11.10
932      3.88  19.64  68.93  63.12  0.00  0.00  0.00  11.19
908      3.85  19.02  69.96  64.29  0.00  0.00  0.00  11.14
906      4.17  21.80  65.17  59.97  0.00  0.00  0.00  11.21
900      4.44  19.55  69.13  64.61  0.00  0.00  0.00  11.14
936      3.91  19.58  69.07  63.52  0.00  0.00  0.00  11.13
908      3.92  19.44  69.93  65.50  0.00  0.00  0.00  11.10
907      4.12  20.46  67.94  63.55  0.00  0.00  0.00  11.16
907      3.96  19.58  68.50  63.36  0.00  0.00  0.00  11.17
1021     4.08  20.86  63.52  53.40  0.00  0.00  0.00  11.41
900      4.09  18.68  70.70  65.87  0.00  0.00  0.00  11.09
^C900     3.79  17.58  72.97  68.44  0.00  0.00  0.00  11.06
root@proxmox:~#
```

- turbostat: GTX 960 removed, pfSense on

```

root@proxmox:~# turbostat --Summary --quiet --show Pkg%pc2,Pkg%pc3,Pkg%pc6,Pkg%pc7,CPU%c1,CPU%c6,CPU%c7,Bzy_MHz,PkgWatt --interval 5
Bzy_MHz CPU%c1 CPU%c6 CPU%c7 Pkg%pc2 Pkg%pc3 Pkg%pc6 Pkg%pc7 PkgWatt
3851 6.05 20.06 71.11 68.81 0.00 0.00 0.00 5.04
3871 5.61 18.99 73.66 72.17 0.00 0.00 0.00 4.47
3850 5.85 19.95 71.79 70.28 0.00 0.00 0.00 4.83
3867 5.66 18.93 73.57 71.94 0.00 0.00 0.00 4.44
3852 5.89 20.14 71.52 70.20 0.00 0.00 0.00 4.83
3862 5.72 18.88 73.33 71.18 0.00 0.00 0.00 4.57
3851 5.86 19.84 71.91 70.19 0.00 0.00 0.00 4.80
3858 5.74 19.02 72.98 70.73 0.00 0.00 0.00 4.73
3859 5.92 19.69 71.42 68.57 0.00 0.00 0.00 5.05
3863 5.76 18.98 73.33 71.14 0.00 0.00 0.00 4.51
3852 5.86 20.01 71.71 70.22 0.00 0.00 0.00 4.79
3877 5.58 18.98 73.71 72.32 0.00 0.00 0.00 4.41
3846 5.95 20.13 71.31 69.63 0.00 0.00 0.00 4.93
3856 5.94 19.33 72.86 71.38 0.00 0.00 0.00 4.55
3827 6.54 20.14 70.30 67.82 0.00 0.00 0.00 5.12
3859 5.96 19.15 72.94 70.92 0.00 0.00 0.00 4.61
3847 6.00 20.07 71.44 69.86 0.00 0.00 0.00 4.85
3855 5.79 18.80 73.32 71.26 0.00 0.00 0.00 4.58
3833 6.37 19.84 70.95 68.89 0.00 0.00 0.00 4.98
3842 6.15 18.69 72.94 70.18 0.00 0.00 0.00 4.67
^C3818 7.02 18.87 68.09 63.19 0.00 0.00 0.00 6.22
root@proxmox:~#

```

- turbostat: GTX 960 removed, pfSense off

```

root@proxmox:~# turbostat --Summary --quiet --show Pkg%pc2,Pkg%pc3,Pkg%pc6,Pkg%pc7,CPU%c1,CPU%c6,CPU%c7,Bzy_MHz,PkgWatt --interval 5
Bzy_MHz CPU%c1 CPU%c6 CPU%c7 Pkg%pc2 Pkg%pc3 Pkg%pc6 Pkg%pc7 PkgWatt
3830 0.68 0.70 98.39 97.17 0.00 0.00 0.00 0.85
3844 1.17 2.46 95.31 93.42 0.00 0.00 0.00 1.52
3853 0.79 0.84 97.97 96.15 0.00 0.00 0.00 0.95
3847 0.96 1.87 96.35 94.52 0.00 0.00 0.00 1.30
3841 0.71 0.71 98.30 96.82 0.00 0.00 0.00 0.90
3854 0.96 1.81 96.41 94.48 0.00 0.00 0.00 1.29
3827 0.74 0.79 98.11 96.58 0.00 0.00 0.00 0.95
3858 1.00 1.69 96.35 93.67 0.00 0.00 0.00 1.39
3841 0.70 0.69 98.35 96.85 0.00 0.00 0.00 0.88
3854 0.95 1.76 96.47 94.51 0.00 0.00 0.00 1.31
3827 0.71 0.79 98.23 96.95 0.00 0.00 0.00 0.88
3858 1.08 1.67 96.28 93.46 0.00 0.00 0.00 1.43
3849 0.67 0.76 98.32 97.02 0.00 0.00 0.00 0.88
3857 1.04 2.14 95.80 93.57 0.00 0.00 0.00 1.47
3839 0.76 0.74 98.21 96.69 0.00 0.00 0.00 0.92
3843 0.97 1.83 96.38 94.58 0.00 0.00 0.00 1.30
3858 0.80 0.63 98.18 95.95 0.00 0.00 0.00 0.99
3846 0.97 1.74 96.48 94.45 0.00 0.00 0.00 1.32
3829 0.71 0.72 98.22 96.70 0.00 0.00 0.00 0.94
3852 0.99 1.76 96.43 94.44 0.00 0.00 0.00 1.32
3859 0.75 0.59 98.28 96.18 0.00 0.00 0.00 0.95
3851 0.97 1.67 96.54 94.44 0.00 0.00 0.00 1.32
3837 0.68 0.71 98.36 96.92 0.00 0.00 0.00 0.87
3854 0.97 1.83 96.38 94.36 0.00 0.00 0.00 1.31
^C3853 0.95 1.43 97.21 95.77 0.00 0.00 0.00 1.06
root@proxmox:~#

```

- PowerTOP overview (pfSense off)

Summary: 457.6 wakeups/second, 0.0 GPU ops/seconds, 0.0 VFS ops/sec and 7.6% CPU use

Usage	Events/s	Category	Description
1.5 ms/s	292.0	Timer	tick_nohz_handler
405.2 μ s/s	32.3	kWork	usb_giveback_urb_bh
286.6 μ s/s	16.7	Process	[PID 15] [rcu_preempt]
0.9 ms/s	13.7	Interrupt	[7] sched(softirq)
74.1 μ s/s	13.7	kWork	delayed_vfree_work
25.4 ms/s	1.0	Process	[PID 1052] pve-firewall
28.8 μ s/s	9.8	kWork	psi_avgs_work
439.5 μ s/s	8.8	kWork	handle_update
295.8 μ s/s	8.8	Process	[PID 949] /usr/bin/python3 /usr/bin/fail2ban-server -xf start
61.2 μ s/s	4.9	Process	[PID 690] /usr/bin/rrdcached -g
55.1 μ s/s	3.9	Process	[PID 565] /usr/sbin/pvefw-logger
19.8 μ s/s	3.9	Timer	watchdog_timer_fn
3.2 ms/s	2.0	Process	[PID 4268] powertop
2.9 ms/s	1.0	Process	[PID 4282] pve-firewall
2.9 ms/s	1.0	Process	[PID 4288] pve-firewall
2.9 ms/s	1.0	Process	[PID 4284] pve-firewall
2.9 ms/s	1.0	Process	[PID 4287] pve-firewall
2.8 ms/s	1.0	Process	[PID 4286] pve-firewall
2.8 ms/s	1.0	Process	[PID 4290] pve-firewall
2.7 ms/s	1.0	Process	[PID 4283] pve-firewall
2.7 ms/s	1.0	Process	[PID 4289] pve-firewall
2.7 ms/s	1.0	Process	[PID 4285] pve-firewall
106.1 μ s/s	2.0	Interrupt	[3] net_rx(softirq)
68.3 μ s/s	2.0	Process	[PID 950] /usr/bin/python3 /usr/bin/fail2ban-server -xf start
40.7 μ s/s	2.0	Process	[PID 2275] /usr/bin/rrdcached -g
24.7 μ s/s	2.0	Process	[PID 1662] /usr/bin/rrdcached -g
10.9 μ s/s	2.0	Process	[PID 51] [kcompactd0]
1.0 ms/s	1.0	Process	[PID 4296] sh
1.0 ms/s	1.0	Process	[PID 4294] sh
0.9 ms/s	1.0	Process	[PID 4293] ebtables-save
0.8 ms/s	1.0	Process	[PID 4297] ebtables-save
0.7 ms/s	1.0	Process	[PID 4292] ebtables-save
584.9 μ s/s	1.0	Process	[PID 4295] sh
2.7 ms/s	0.00	Process	[PID 4298] pve-firewall
215.5 μ s/s	1.0	Process	[PID 1094] pveproxy worker
2.7 ms/s	0.00	Process	[PID 4291] pve-firewall
166.6 μ s/s	1.0	Process	[PID 1028] /usr/bin/pmxdfs
61.2 μ s/s	1.0	Process	[PID 1083] pvedaemon worker